Subject Code : 23A01703a

GROUND IMPROVEMENT TECHNIQUES

UNIT - III

Densification Methods in Granular and Cohesive Soils

Syllabus topics covered in this unit

- In-situ Densification in Granular Soils: Vibration at Ground Surface, Impact at Ground Surface
- Vibration at Depth, Impact at Depth
- In-situ Densification in Cohesive Soils: Preloading or Dewatering
- Vertical Drains: Sand Drains, Sand Wick, Geodrains
- Stone and Lime Columns
- Thermal Methods

How to use these notes

- Red words = keywords to remember for the exam.
- Green words = examples, units and values.
- Yellow boxes = quick points; pink boxes = important questions / formulae.
- Learn the sketches - they fetch marks in the answer paper.

1. Need for Densification and Classification of Methods

Densification (compaction in place) increases the density of loose soil, which increases shear strength and bearing capacity, reduces settlement and compressibility, permeability, and the danger of liquefaction. The method depends on the soil type.

- Granular soils (sand, gravel; free draining): densified by rearranging the grains into a denser state by vibration or impact - the soil volume reduces immediately.
- Cohesive soils (clay, silt; low permeability): the density is raised only by squeezing the water out (consolidation) over time, so they are treated by preloading, drains, columns, and heat.

Soil	Principle	Methods
Granular (cohesionless)	Particle rearrangement by vibration / impact; relative density D_r increases	Surface: vibratory roller, plate vibrator, dynamic compaction (drop weight at the surface). Depth: vibro-compaction (vibroflotation), compaction piles, blasting, deep dynamic compaction
Cohesive (fine grained)	Expulsion of water (consolidation), replacement or reinforcement, chemical change	Preloading and surcharge, vertical drains (sand drain, wick drain), stone columns, lime columns, electro-osmosis, thermal methods

Relative density and compaction

- $D_r = (e_{max} - e) / (e_{max} - e_{min}) \times 100$ percent; D_r < 15 very loose, 15 - 35 loose, 35 - 65 medium, 65 - 85 dense, > 85 very dense.
- Targets: raise D_r to 70 - 85 percent (SPT N > 20 - 30) for foundations on sand and to avoid liquefaction.

Depth of treatment and soil suitability

Method	Depth of treatment	Soil
Vibratory roller / plate	0.5 - 2 m	Clean sand, gravel
Dynamic compaction (DC)	5 - 12 m (deeper with a heavy weight)	Granular, fill, waste, partly cohesive
Vibro-compaction	10 - 30 m	Clean sands with fines < 10 - 15 %
Compaction piles	10 - 20 m	Loose sands
Blasting	10 - 20 m	Saturated loose clean sand
Preloading, vertical drains	10 - 30 m	Soft clay, silt
Stone columns	6 - 20 m	Soft clay, silt, loose sand

2. Granular Soils: Vibration and Impact at the Ground Surface

A. Vibration at the ground surface

The surface of loose granular soil is compacted by vibrating equipment. The vibration reduces the inter-particle friction temporarily, so that the grains slide into a denser state under their own weight and the machine weight.

- Vibratory smooth drum roller (self propelled, 8 - 12 tonnes): frequency 25 - 50 Hz, amplitude 0.5 - 2 mm; 4 - 8 passes for lifts of 0.3 - 0.6 m. Best for sands and gravels.
- Vibrating plate compactor (plate vibrator) and vibratory tamper (rammer) for small areas, trenches, backfills; vibrating tyre rollers, vibrating pad foot rollers.
- Frequency should be close to the natural frequency of the soil (resonance) - sand 20 - 40 Hz; amplitude 1 - 2 mm; speed of travel 2 - 4 km/h; thickness of the layer 0.3 - 1.5 m (loose).
- Control: field density by sand replacement or core cutter, nuclear gauge; target 95 - 100 percent of the standard / modified Proctor dry density or the relative density 70 - 75 percent.
- Limitations: shallow effect (about 1 m), not for clay or silt (damping). Used for the base of the pavement, backfill, embankment, and as prior compaction of the sand fills.

B. Impact at the ground surface

- Rammers and impact rollers: heavy rammers, drop hammers and non-circular rollers (square, 3-sided impact rollers of 8 - 12 tonnes) deliver the impact energy deep (up to 2 - 3 m).
- Dynamic compaction (DC) (Menard, 1970 - France): a heavy pounder of 10 - 40 tonnes (steel or concrete block) is dropped repeatedly from 10 - 40 m height by a crane in a grid pattern on the

ground surface. Impact generates shock waves (compression, shear and Rayleigh waves) that densify the loose granular soil, collapse the voids, liquefy the soil momentarily and help to re-arrange the particles.

Depth of influence (Menard)

- $D = n \sqrt{(W H)}$ (D in m; W = weight of pounder in tonnes; H = drop height in m)
- n = 0.3 - 0.8 (0.5 typical for sand; 0.35 - 0.4 for silty, clay-fill); Example: W = 15 t, H = 20 m => $D = 0.5 \times \sqrt{300} = 8.7$ m.
- Energy per blow = $W H$; energy per unit area: 1000 to 2500 kJ/m^2 for the main pass (compare Proctor 600 kJ/m^3).

Primary grid (large spacing) - high energy → Rest for the pore pressure to dissipate → Secondary / intermediate grid → Ironing pass - low energy to compact the top

- Procedure: 2 - 3 passes; grid spacing 1.5 - 2.5 times the diameter of the pounder (4 - 10 m), 5 - 15 drops per point; the craters (1 - 2 m deep) are filled with sand or gravel after each pass; the ironing pass compacts the top 2 m using low drop.
- Waiting time: days to weeks for the pore water pressure to dissipate in the sands with fines or saturated soil (0 for the clean sand).
- Applications: loose sand, fill, landfills, collapsible soil, reclaimed land (airports - Kansai, Chennai, Mumbai) and liquefiable sites, industrial yards; rapid impact compaction (RIC) with 7 t weight, 1.2 m drop and the high frequency (40 blows/min) is a modified type.
- Precaution: ground vibration and noise (safe distance > 30 m from the structures - peak particle velocity < 5 mm/s); flying stones; the water table should be > 2 m below the surface (or a working mat of 0.5 - 1 m gravel is placed).

3. Granular Soils: Vibration at Depth

The probes or vibrators are inserted to the depth of the loose soil and densify it from within, giving uniform improvement with depth up to 30 m.

Vibro-compaction (Vibroflotation)

- Developed by Steuerman (1930s, Germany); the vibroflot is a cylinder (30 - 45 cm in diameter, 2 - 5 m long, 1.5 - 4 t) containing an eccentric weight rotated by an electric or hydraulic motor,

producing horizontal vibration (frequency 30 - 50 Hz, amplitude 5 - 12 mm). Water jets at the tip and the sides (jetting) help penetration.

- Procedure: (1) the vibrator is lowered with water jets to the design depth; (2) the vibration at the depth densifies the soil in a circle around it (radius 1.5 - 3 m) - the soil is liquefied and then it re-settles in a dense state; (3) the vibrator is lifted in stages of 0.3 - 0.5 m while the sand (backfill) is added at the top to fill the crater formed by the settlement; (4) the process is repeated at points in a triangular or square grid of 1.5 - 3 m.
- Dry bottom feed (vibro-replacement) and wet top feed methods are used.
- Suitability: clean sands and gravels with less than 10 - 15 percent fines (< 0.075 mm) and less than 2 percent clay; the grain size band of the Brown (1977) suitability number: $SN = 1.7 \sqrt{3/D_{50}^2 + 1/D_{20}^2 + 1/D_{10}^2}$ - excellent for SN 0 - 10, good 10 - 20, fair 20 - 30, poor 30 - 50, unsuitable > 50.
- Result: relative density increases to 70 - 85 percent; the SPT N value increases 2 - 3 times; the settlement reduced to 25 mm or less; liquefaction resistance increased. Deep treatment 10 - 20 m (up to 30 m).

Terra-probe and vibro-rods

- Terra-probe: a large steel pipe (75 cm dia, 20 m) with a vibratory hammer at the top (15 Hz) is driven (vibrated) into the sand and withdrawn at 0.3 m / 10 s; compaction in the zone of 2 - 3 m around.
- Vibro-rod / vibro-wing: a steel H-section or pipe with wings vibrated into the soil - for deep densifying of 10 - 20 m by the vibrating hammer of 15 Hz; used in the sand, cheap but a lower density increase than the vibroflot.
- Vibro-replacement / stone columns: in the sand with a high fines content the vibrator forms the hole, which is filled with crushed stone and compacted - this is the vibro-replacement (see the stone column).

Aspect	Vibro-compaction	Vibro-replacement (stone column)
Soil	Clean sand and gravel (fines < 10 %)	Silty sand, clay, soft soil
Mechanism	Densifies the existing soil	Replaces the soil with the compacted stone and reinforces
Backfill	Sand (or none)	Crushed stone 20 - 75 mm
Spacing	1.5 - 3 m	1.5 - 3 m

4. Granular Soils: Impact at Depth

Compaction piles (Compozer, sand compaction piles)

- A hollow steel casing (30 - 60 cm) is driven by the vibratory hammer into the loose soil to the depth. Sand (or gravel / crushed stone) is fed through the casing; the casing is repeatedly pulled up and re-driven (by vibration or driving) so that the sand is compacted and displaced into the soil, forming a dense sand pile of larger diameter (0.7 - 1.0 m) and densifies the surrounding soil by displacement.
- Spacing 1.5 - 2.5 m in a triangular grid; the area replacement ratio $a_s = A_{pile} / A_{tributary} = 0.1 - 0.4$ and the soil improvement is by volume displacement.
- Suitable for: loose sands, silty sand, soft clay (as a Japanese practice - sand compaction pile SCP - provides drainage + reinforcement), 10 - 30 m depth (offshore also). Not suitable when the vibration is a problem (noise, vibration to the nearby structures).
- Improvement: SPT N rises by 2 - 4 times; the bearing capacity of the composite ground improved; settlement and liquefaction reduced.

Blasting (explosive compaction)

- Explosive charges (1 - 5 kg of dynamite or TNT) are buried at the depth (5 - 20 m) in the saturated loose clean sand and detonated in the pattern. The shock wave and the sudden rise of the pore pressure cause liquefaction and then resettlement to the densified state: the settlement of 2 - 10 % of the depth.
- Several (3 - 5) explosions are repeated in a series of detonations at intervals. Cost low per m^3, quick; used for large areas, remote sites, old fill.
- Precaution: ground vibration, structures nearby; may not be used within 100 - 200 m of buildings, requires a licence and control; the effect uncertain; difficult for the controlled quality.

Deep (heavy) dynamic compaction and other impact methods

- Deep dynamic compaction with the heavy pounder 40 - 200 tonnes dropped from 25 - 40 m reaches a depth of 20 - 30 m (the Menard rule $D = n \sqrt{W H}$); also Rapid Impact Compactors, hydraulic hammer driving steel pipes / rams (Franki piles - compaction).
- Impact at depth by driven piles: displacement piles (timber, precast) installed at close spacing (compaction piles) which densify the soil between them; with the pile spacing 3 - 5 diameters.
- Vibro-driven stone columns, Deep soil mixing and jet grouting are other deep methods but they are the soil stabilisation, not the pure densification.

Method	Energy source	Depth	Remarks
Vibroflotation	Rotating eccentric (horizontal vibration)	10 - 30 m	Clean sand
Terra-probe	Vertical vibration of pipe	10 - 20 m	Cheap, slow
Compaction piles	Vibro / impact driving + sand	10 - 30 m	Displacement
Blasting	Explosives	5 - 20 m	Loose saturated sand
Dynamic compaction	Falling weight	5 - 30 m	Wide soil range

Quality control of the densification

- Before and after: SPT, CPT, DMT, shear wave velocity, plate load test; target $N_{1,60}$ > 20 - 25 (liquefaction resistant), q_c > 10 - 12 MPa, D_r 70 - 80 %.
- Check the settlement of the surface (crater volume), ground heave, lateral movements; pore pressure with piezometers to fix the rest period.

5. Cohesive Soils: Preloading or Dewatering

In the saturated clay the density increases only when the pore water drains out under a higher effective stress (consolidation). The process is slow because k is very low (10^{-7} to 10^{-9} cm/s). Preloading applies the future load (or more) before construction so that most of the primary consolidation settlement takes place beforehand.

Preloading (surcharge) method

Place the fill (surcharge) over the site → Pore water squeezed out - consolidation → Monitor settlement and pore pressure → Remove the surcharge, then construct

- A fill (earth, sand, stockpile, or water tank loaded) of height equal to or more than the structure load is placed. Surcharge (extra fill) is used to reduce the time: surcharge ratio R = Delta $sigma_s$ / Delta $sigma_f$ (typical 1.2 - 1.5 or more).
- Time of consolidation: $t = T_v H_{dr}^2 / c_v$; for U = 90 %, T_v = 0.848. The time for the soft clay of thickness 10 m (one-way drainage, $c_v = 1 \times 10^{-3}$ cm^2/s) is long: $0.848 \times (1000)^2 / 10^{-3} = 8.5 \times 10^8$ s = 27 years! Therefore vertical drains are used to shorten the time.
- Settlement: $S_c = (C_c H / (1 + e_0)) \log ((sigma_0' + Delta\ sigma) / sigma_0')$.

- With the surcharge, the soil reaches the degree of consolidation required in the shorter time under the larger load; remove the surcharge when the settlement under the surcharge equals the final settlement under the permanent load (i.e., U_{design} achieved by the Asaoka / hyperbolic method of monitoring).
- Instrumentation: settlement plates, piezometers, inclinometers, extensometers; the construction of the fill in the stages to avoid bearing failure (strength gain by consolidation; Skempton's method).

Example

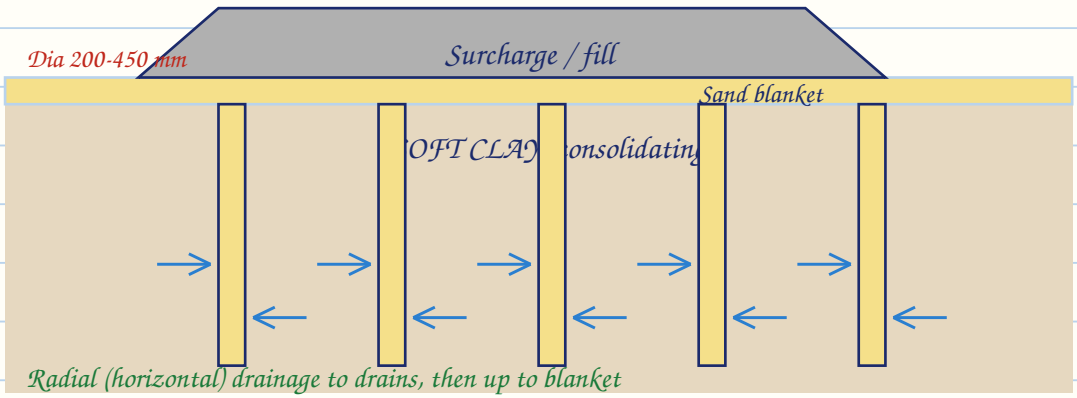
- Final settlement 0.5 m under the load p; a surcharge of 1.5 p is placed. Desired 90 % of the 0.5 m = 0.45 m.
- The surcharge causes S = 0.45 m at a U_s of 60 % of its (larger) ultimate settlement - the time required is only for U about 60 % (T_v = 0.287) instead of 90 % (T_v = 0.848) - time cut to 1/3.

Dewatering as a preload

- Lowering the ground water table by pumping (well points, deep wells) raises the effective stress by $gamma_w$ x (drop in the water table): an equivalent load of 10 kPa for each 1 m lowering. Used in the sand and silt layers and to preload by the vacuum.
- Vacuum preloading (Kjellman, 1952): an airtight membrane (PVC / LDPE film) is laid over the sand blanket, with the vertical drains connected to the peripheral trench with the vacuum pump. A vacuum of -70 to -90 kPa gives the pre-compression equal to 70 - 90 kPa of the surcharge (no risk of the bearing failure, since the confining stress does not rise), and the lateral deflection is inward. Combined with the fill: vacuum + surcharge.
- Electro-osmosis (covered in unit II) is also a dewatering method for the clay.
- Applications: highways, embankments on the soft clay (e.g., Ennore, Mumbai - Navi Mumbai airport), port and airport reclamation, storage tanks, large warehouse floors.

6. Vertical Drains - Sand Drains, Sand Wicks, Geodrains

To shorten the consolidation time, vertical drains are installed in the soft clay so that the drainage path is reduced from the layer thickness H to the horizontal radial distance (half the drain spacing, 1 - 2 m). The time of consolidation is reduced by 10 - 100 times.



Sand drains

- Vertical holes (dia 200 - 450 mm, up to 0.45 m) bored or driven in a triangular or square grid at spacing 1.5 - 4 m and filled with the clean coarse sand ($k > 10^{-2}$ cm/s, which should be at least 100 times that of the clay). Top is covered by the sand blanket (0.5 - 1 m) that carries the water away.
- Installation: displacement (driven mandrel - disturbs the clay and the smear), non-displacement (augered / jetted), the Menard and Raymond methods; the effective diameter of the drain zone d_e = 1.05 s (triangular) or 1.13 s (square).
- Limitations: smear zone, discontinuity (necking) of the sand, large settlements can cut the sand column, costly, and the sand drain needs the large quantity of sand.

Prefabricated vertical drains (PVD) - sand wicks and geodrains

- Sand wick: a sand-filled geotextile sock (small diameter 65 - 100 mm); sand drain with the filter wrap - inexpensive; better than the sand drains but higher costs and the possible bending.
- Geodrains (band drains, wick drains, PVDs): a plastic core (polypropylene, polyethylene) with grooved channels covered by a non-woven geotextile filter jacket; band 100 mm wide and 3 - 6 mm thick; equivalent diameter $d_w = (a + b) / 2$ (a = width, b = thickness), about 50 - 70 mm. Examples: Mebra drain, Colbond, Alidrain, Geodrain, Bonded fibre drain (Kjellman cardboard wick drain, 1948).
- Installation: the drain is fed through a hollow steel mandrel, pushed into the ground with the anchor plate (static push or vibratory) and cut. Speed: up to 1 m/s, 10 - 40 m deep, 1500 m of drain/h per rig. The spacing is 1 - 2 m.
- Advantages of PVD: fast, cheap, continuous, little smear and disturbance, quality assured at the factory, no sand.

Design - Barron / Hansbo radial consolidation

Design equations

- Overall degree: $(1 - U) = (1 - U_v)(1 - U_r)$ (Carrillo)
- Radial: $U_r = 1 - \exp(-8 T_r / mu)$ with $T_r = c_h t / d_e^2$
- Hansbo: $t = (d_e^2 / (8 c_h))[\ln (d_e/d_w) + (k_h/k_s) \ln s - 0.75] \ln (1 / (1 - U_r))$
- Spacing ratio $n = d_e / d_w$; $mu = (n^2/(n^2 - 1)) \ln n - (3 n^2 - 1)/(4 n^2)$ (Barron, ideal drain, no smear).
- $c_h = (2 - 5) \times c_v$ for the clay; drain spacing S chosen to give U about 90 % in the required time (3 - 6 months).

- Well resistance and smear effect ($k_s < k_h$, the disturbed zone $d_s = 2 d_m$) are accounted for; the capacity of the drain q_w should be > 100 m^3/year.
- For the depth, the drains penetrate the full compressible layer or a part, at least up to the critical depth for the stability.

7. Stone and Lime Columns

Stone columns (vibro-replacement)

- Stone columns are the vertical columns of the compacted crushed stone or gravel (dia 0.6 - 1.0 m, length 3 - 15 m, spacing 1.5 - 3 m) installed in soft clays, silts and loose silty sands by the vibroflot (vibro-replacement, wet top feed or dry bottom feed) or by the ram (Franki) or the cased auger (Rammed stone column).
- Functions: (i) reinforcement - the stiff columns carry the larger load (stress concentration ratio n = $sigma_s / sigma_c$ = 3 - 6); (ii) densification of the surrounding loose soil; (iii) drainage - faster consolidation (and the dissipation of the excess pore pressure in earthquakes); (iv) reduction of settlement by 50 - 70 % and the increase in the bearing capacity two to three times; (v) improves the stability of slopes.
- Stone: 20 - 75 mm, hard, angular, durable, clean. Condition: the undrained shear strength of the clay c_u > 15 - 20 kPa (to give the lateral support to the column); for c_u < 15 kPa, the column is encased in the geosynthetic.

Design of the stone column (Priebe / Greenwood)

- *Area replacement ratio $a_s = A_c / A$ ($A = 0.866\ s^2$ for triangular, s^2 for the square grid; $A_c = pi\ d^2/4$).*
- *Ultimate capacity of the column (Hughes and Withers): $q_{ult} = sigma_{r,max}\ (1 + sin\ phi_c) / (1 - sin\ phi_c) = K_p\ sigma_{r,max}$ with $sigma_{r,max} = sigma_{r0} + 4\ c_u$ (bulging limit).*
- *Typically phi_c = 38 - 42 degree: q_{ult} about 400 - 800 kPa, safe load 100 - 400 kN per column.*
- *Priebe improvement factor $n_0 = 1 + a_s\ [\ 0.5 + f(mu_s, a_s)\ ] / (\ K_{ac}\ f(mu_s, a_s)\ ) - 1$; Settlement of improved ground $S = S_0 / n_0$.*

Lime columns and lime piles

- *Lime column (lime pile): holes (dia 0.2 - 0.5 m, depth 3 - 10 m) bored in the soft clay are filled with quick lime (CaO) or lime + sand / fly ash and compacted - used since the Roman times. Lime absorbs water from the clay (hydration: $CaO + H_2O \rightarrow Ca(OH)_2$, heat generated, volume increases 2 times), the cation exchange and the pozzolanic reaction increase the strength of the zone around.*
- *Deep lime-cement mixing (DLM / DCM): lime or lime-cement is mixed in-situ with the soil by a rotating mixing tool - the Swedish Lime-cement columns (Broms, 1970s) with dia 0.5 - 0.8 m, depth to 25 m, strength 100 - 500 kPa; used for the highway, rail embankments in the soft clay.*
- *Lime pile effect: reduction of the water content 5 - 10 %, rise in the shear strength 2 - 4 times, the soil near the pile ~ 0.3 - 0.4 m radius becomes the stabilised; improvement is mostly within 3 - 4 diameters; suitable for expansive soils (unit I) as well as soft clays and organic soil.*
- *Quantity of lime: 2 - 6 percent of the soil weight (about 10 - 20 kg per m of column).*

Aspect	Stone column	Lime column
Material	Crushed stone, gravel	Quick lime, lime-cement
Action	Reinforcement, drainage, densification	Chemical stabilisation, drying
Soil	Soft clay, silt, loose sand	Soft / expansive clay, organic
Strength gain	Immediately	With time (curing 28 days)
Settlement	Reduces by 50 - 70 %	Reduces; also controls swelling

8. Thermal Methods

Thermal methods change the soil by heating (or freezing) it to improve the strength and reduce the compressibility and volume change. They are special and costly methods used in unique situations.

1. Heating (thermal stabilisation)

- *The soil is heated by burning fuel in boreholes (hot gas at 100 - 1000 C) in a series of the bores (dia 0.1 - 0.2 m, spaced 2 - 3 m) which are sealed at the top, or by the electrical heaters (conductive heating, in situ vitrification).*
- *Effects: at 100 C the pore water evaporates, the soil dries, the soil volume decreases; at 400 - 600 C the soil particles are irreversibly bonded (the clay minerals lose the adsorbed and lattice water and form a hard material, which no longer swells); at > 1000 C the fusion (vitrification) forms glass-like mass (brick like strength, the "fired" soil). The permeability increases (the soil is cracked), the compressibility reduces, the strength gain 10 - 100 times.*
- *Applications: loess and the collapsible soils, expansive clays, the soft clays around the borehole to form the stabilised soil columns (the thermal piles / "burned" columns used in Romania, Russia and Japan), and landfill / the contaminated soil (the thermal desorption of the organic pollutants).*
- *Limitations: very high energy cost, fuel, the effect limited to about 1 - 2 m around the hole, the safety, and the environment.*

2. Ground freezing

- *The soil moisture is frozen by circulating brine (calcium chloride) at -25 to -35 C or liquid nitrogen (-196 C) through the freeze pipes, which makes the ground an ice-bonded, strong and water proof wall (strength 2 - 10 MPa).*
- *Used as the temporary support for the deep excavations, shafts and tunnels in the saturated soil, to cut-off the water without the need of dewatering, the underpinning, and the cut-off around the contaminated sites. The freeze time is days to months; frost heave (9 % volume expansion) of the soil and the thaw settlement need the care.*

Aspect	Heating	Freezing
Temperature	100 - 1000 C	-25 to -196 C
Result	Permanent improvement (hardening)	Temporary support and cut-off
Soils	Clays, loess, collapsible soils	All saturated soils
Cost	Very high	High

Remember

- Granular soil: vibration + impact (densify the grains). Cohesive soil: drain the water (preloading, drains), replace / reinforce (stone, lime columns), heat (thermal).
- Preloading + vertical drains is the most used method on the soft clay; PVD spacing 1 - 2 m, surcharge 1.2 - 1.5 times the load.

Quick revision of Unit III

- Granular: vibroflotation (clean sand, 1.5 - 3 m grid), dynamic compaction $D = n\sqrt{WH}$, compaction piles, blasting, terra-probe.
- Cohesive: preloading + surcharge, vacuum preloading (-80 kPa), sand drains (200 - 450 mm, 1.5 - 4 m), PVD (band 100 x 4 mm), stone column (stress concentration 3 - 6), lime column, thermal (400 - 1000 C), freezing.
- Hansbo / Barron design for the radial consolidation; $(1 - U) = (1 - U_v)(1 - U_r)$.

Important Questions (JNTUA pattern)

- Explain the in-situ densification methods in granular soils by vibration at the ground surface.
- Explain dynamic compaction. How is the depth of improvement estimated?
- Describe the vibro-compaction (vibroflotation) with a neat sketch. Which soils are suitable?
- Explain the compaction piles and blasting methods of densification.
- Explain the preloading method. How can the time of consolidation be reduced?
- Describe the vertical drains - sand drains, sand wick and geodrains - with their design basis.
- Explain the construction, function and design of stone columns. Compare with lime columns.
- Write short notes on thermal methods of ground improvement.